Jeremy Tan

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Education

Imperial College London — London, UK
MEng Electronic & Information Engineering — *Oct 2023 – Jun 2027*

- Third Year: 81.5%, Dean's List (Top 5% of the year)
- Master's Thesis: *Tiny Concept-PFNs: Compact Bayesian Inference for Reliable Decisions*, supervised by Dr Sonali Parbhoo

Experience

Digital Verification Intern — Apr 2026 – Sep 2026
Qualcomm — *Cambridge, UK*

- Built an MCP (Model Context Protocol) server that lets LLM agents query binary waveform databases directly, speeding up hardware debug.
- Piloted PyUVM and cocotb, Python test frameworks for hardware, to reduce testbench development time.
- Verified low power radio subsystems for wearable chips using UVM.

Hardware Engineering Intern — Jul 2025 – Sep 2025
Infineon Technologies — *Penang, Malaysia*

- Developed SystemVerilog models of analog IPs (SAR ADC, DC–DC converter, thermal sensor, bandgap reference) to speed up mixed-signal simulation and verification.

Undergraduate Research Assistant — Jul 2024 – Sep 2024
Imperial College London — *London, UK*

- Assisted development of LQR, MRAC, and neural network quadcopter controllers in MATLAB/Simulink, integrated with ROS and Gazebo for simulation testing.

Projects

Sparse Matrix Multiplication Accelerator | *SystemVerilog, Python, TSMC 65 nm* — Oct 2026 – Present

- Designing a sparse matrix multiplication (SpMM) accelerator targeting tapeout on a 1 mm^2 TSMC 65 nm die.

Piano Synthesiser | *C++, FreeRTOS, STM32* — Feb 2026 – Mar 2026

- Built a polyphonic piano synthesiser with FreeRTOS task scheduling verified by rate monotonic analysis, and CAN bus messaging between boards.

PintOS (Toy Operating System) | *C, QEMU, GDB* — Jul 2025 – Sep 2025

- Implemented an MLFQS scheduler and user process support: stack argument passing, 13 system calls covering process control and file I/O, and user pointer validation that kills faulting processes without leaking resources.

FPGA Accelerated Fluid Simulation | *Python, C++, CUDA, SystemVerilog* — May 2025 – Jun 2025

- Implemented the Lattice Boltzmann Method in Python, C++, CUDA, and SystemVerilog, benchmarking performance across CPU, GPU, and FPGA.
- Won **Best Second Year Group Project**

Totally Not Mario Party | *React, Three.js, Python, AWS EC2, WebSockets* — Jan 2025 – Mar 2025

- Built a multiplayer 3D party game in a team of 5, where each player uses an FPGA board as a controller; boards stream input over TCP to an asyncio server on AWS EC2, which relays it to browsers over WebSockets.
- Wrote the React Three Fiber game client (player models and animations, jump and collision mechanics, two minigames, scoreboard) and added a Flask and SQLite score API to the server.

C to RISC-V Compiler | *C++, Flex, Bison* — Jan 2025 – Mar 2025

- Built a compiler from preprocessed C90 to RISC-V assembly in C++ with Flex and Bison, covering lexing, parsing, semantic analysis, and code generation.
- Ranked in the top decile of the cohort; passed all public tests and 80% of unseen tests.

Technical Skills

Languages: Python, C, C++, CUDA, TypeScript/JavaScript, SQL
Frameworks: React, Node.js, Express, FastAPI, PyTorch, NumPy, FreeRTOS, ROS
Tools: Git, Docker, Jenkins, PostgreSQL, GDB, QEMU
Hardware: SystemVerilog, UVM, SVA, cocotb, PyUVM, Verilator, Vivado, MATLAB/Simulink
Spoken Languages: English, Mandarin, Malay